3 In an internal combustion (i.c.) engine, there are cylinders with movable pistons. Each cylinder contains a gas which can be treated as a monatomic ideal gas.

During the thermodynamic process of the i.c. engine called compression stroke, the gas in a cylinder is compressed rapidly from an initial volume of 6.0×10^{-4} m^3 to a final volume of 5.0×10^{-5} m^3. The initial temperature and pressure of the gas was 300 K and 1.0×10^5 Pa respectively.

(a) Calculate the amount, in mol, of the gas in the cylinder.

amount = ... mol [2]

(b) At the end of the compression stroke, the gas reaches a temperature of 550 K.

Determine the pressure of the gas.

pressure = ... Pa [2]

(c) Calculate the increase in internal energy of the gas due to this compression.

increase in internal energy = ... J [2]

(d) (i) State the first law of thermodynamics.

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..... [1]

(ii) The efficiency of the i.c. engine depends on the temperature difference between the start and end of the compression stroke. The greater the temperature difference, the higher the efficiency of the engine.

Use the first law of thermodynamics to explain why the compression stroke is designed to take place rapidly for higher efficiency.

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..... [3]

[Total: 10]